



Filter Wizard

Filter Wizard Design

Created on 06/09/2025



Filter Wizard Design Report

Filter Requirements for Low-Pass, 10th order Butterworth

Specifications: Optimize: Power; +Vs: 5; -Vs: -5

Gain: 0 dB

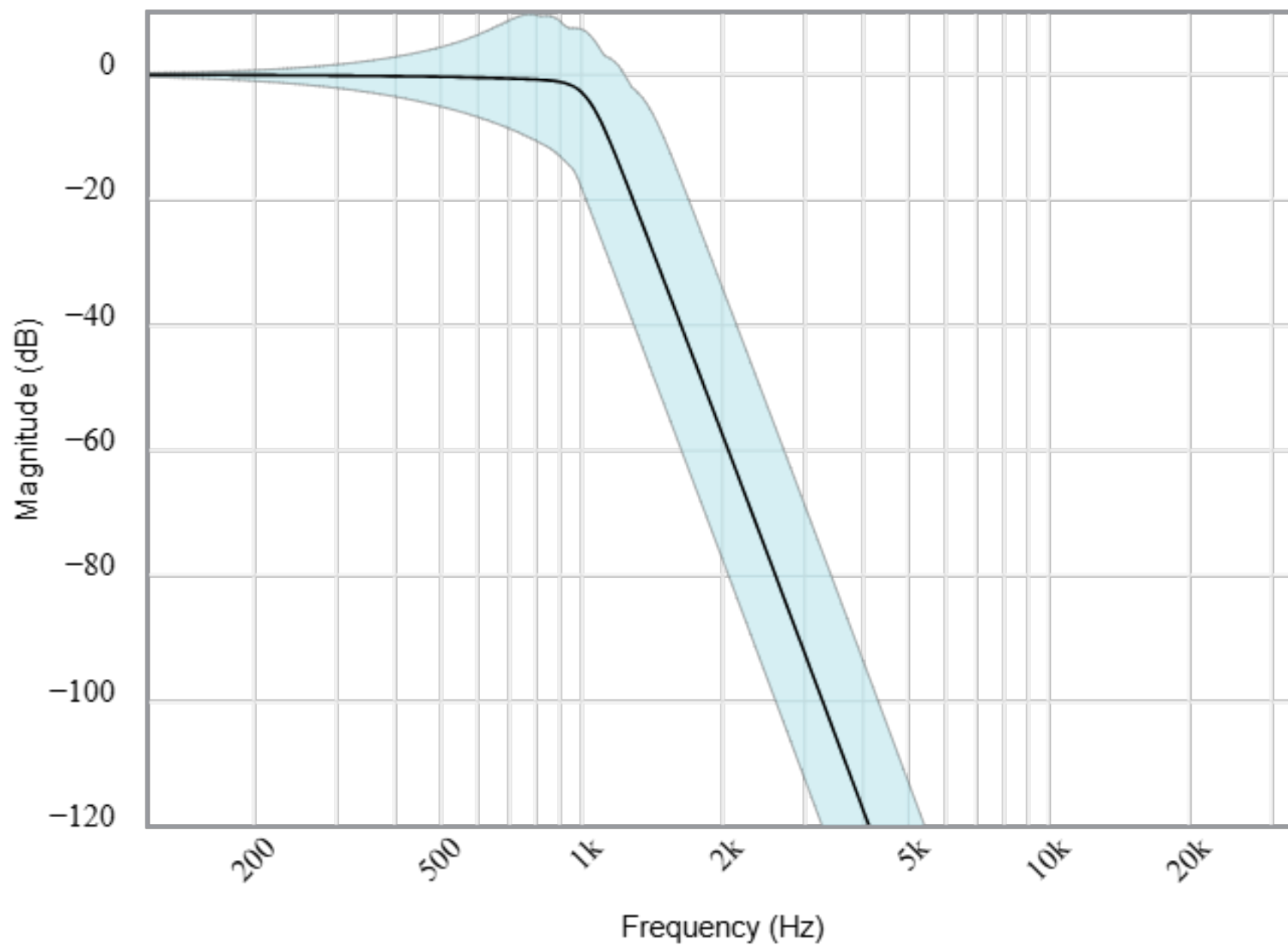
Passband: -10dB at 1.2kHz

Stopband: -100dB at 3.4kHz

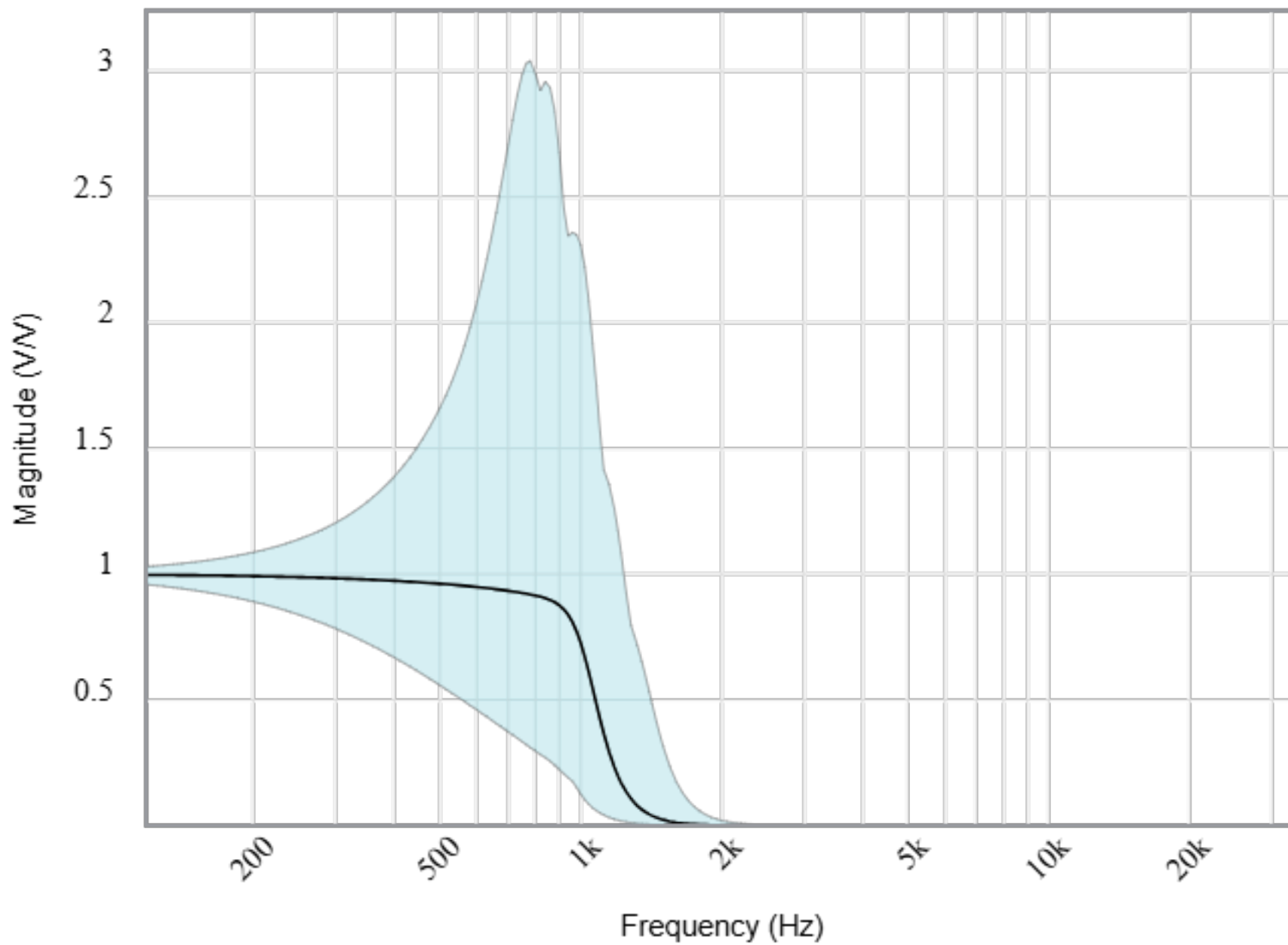
Component Tolerances: Capacitor = 20%; Resistor = 5%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

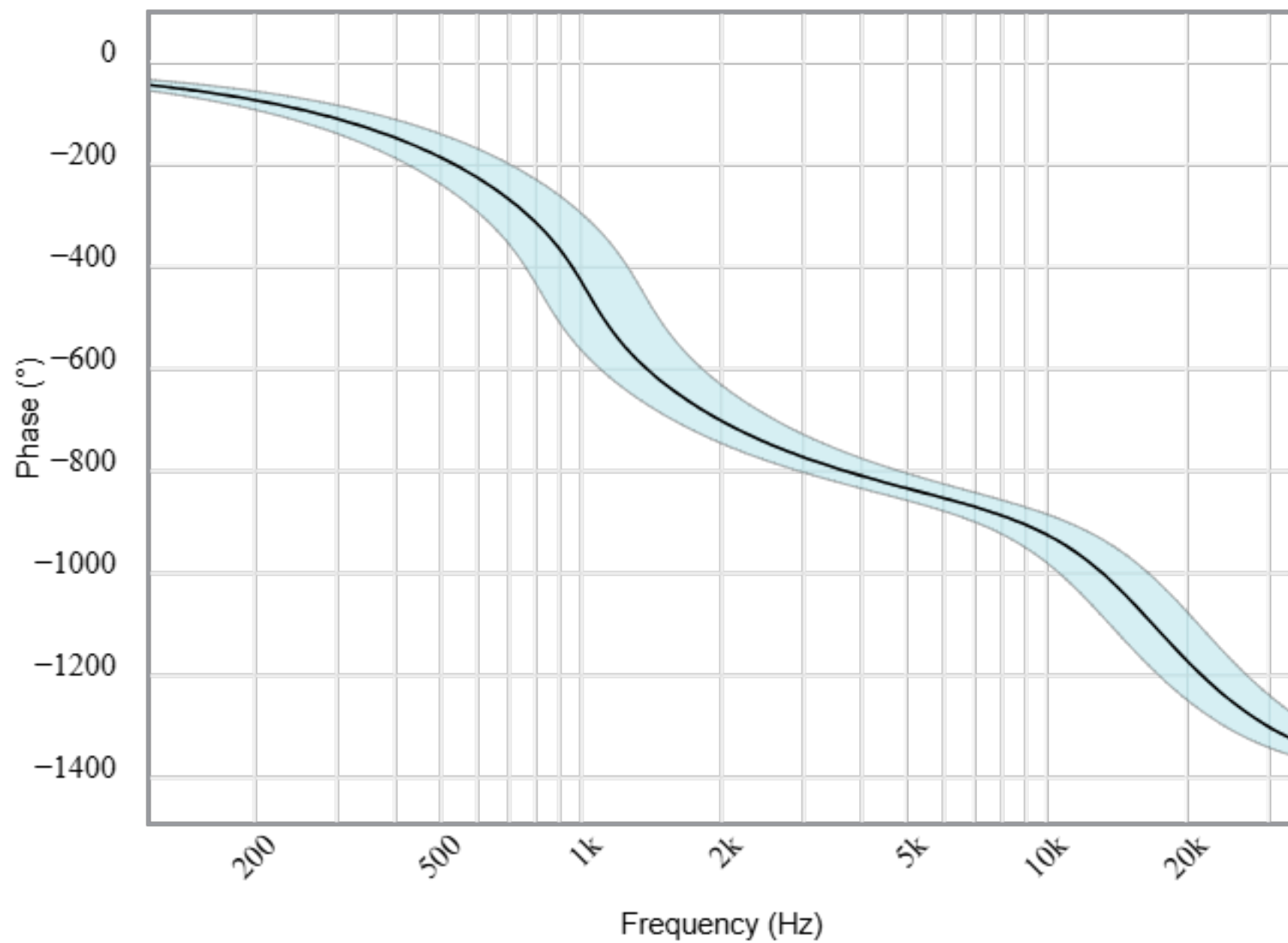
Magnitude(dB)



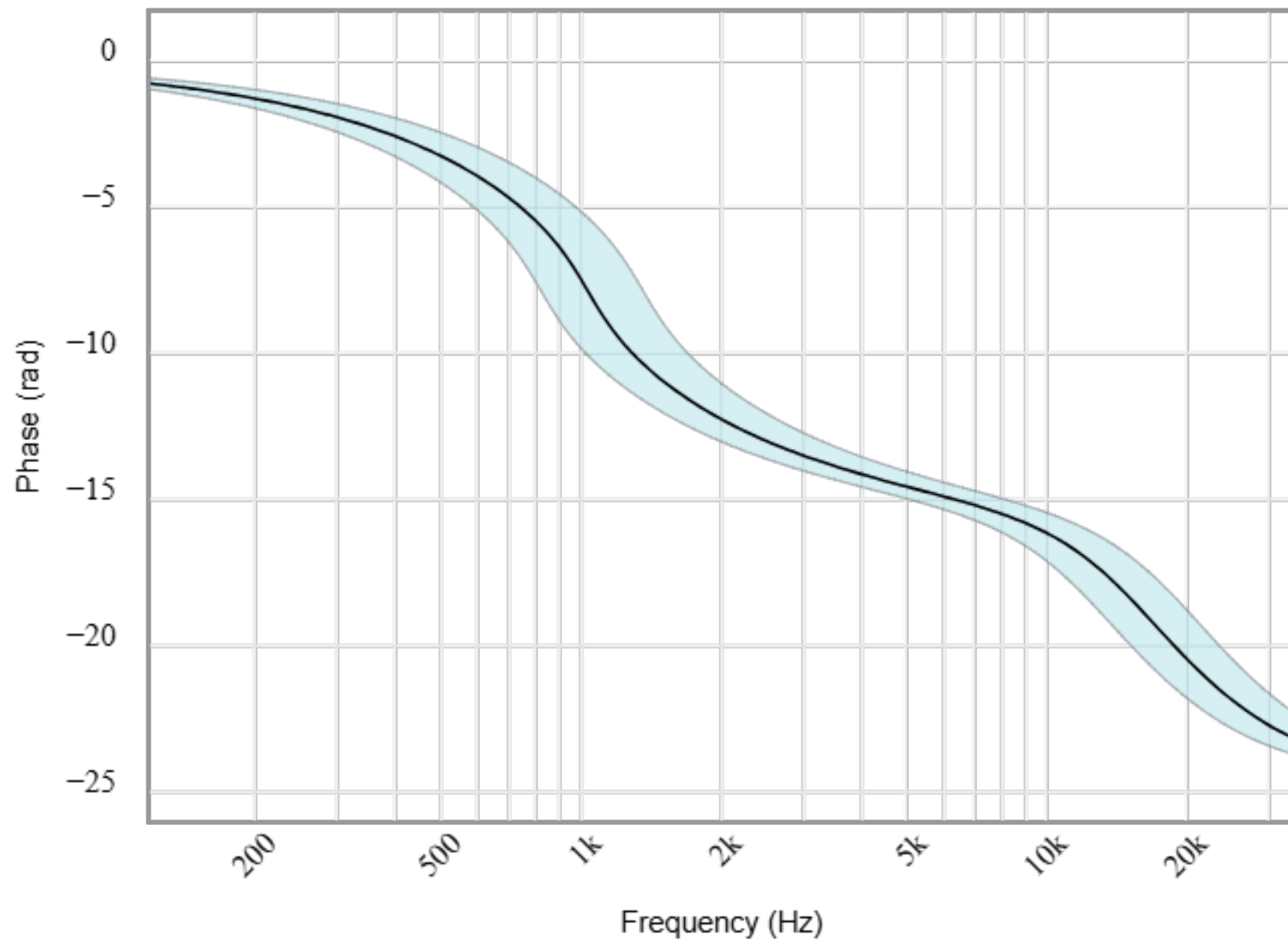
Magnitude(Volts per Volt)



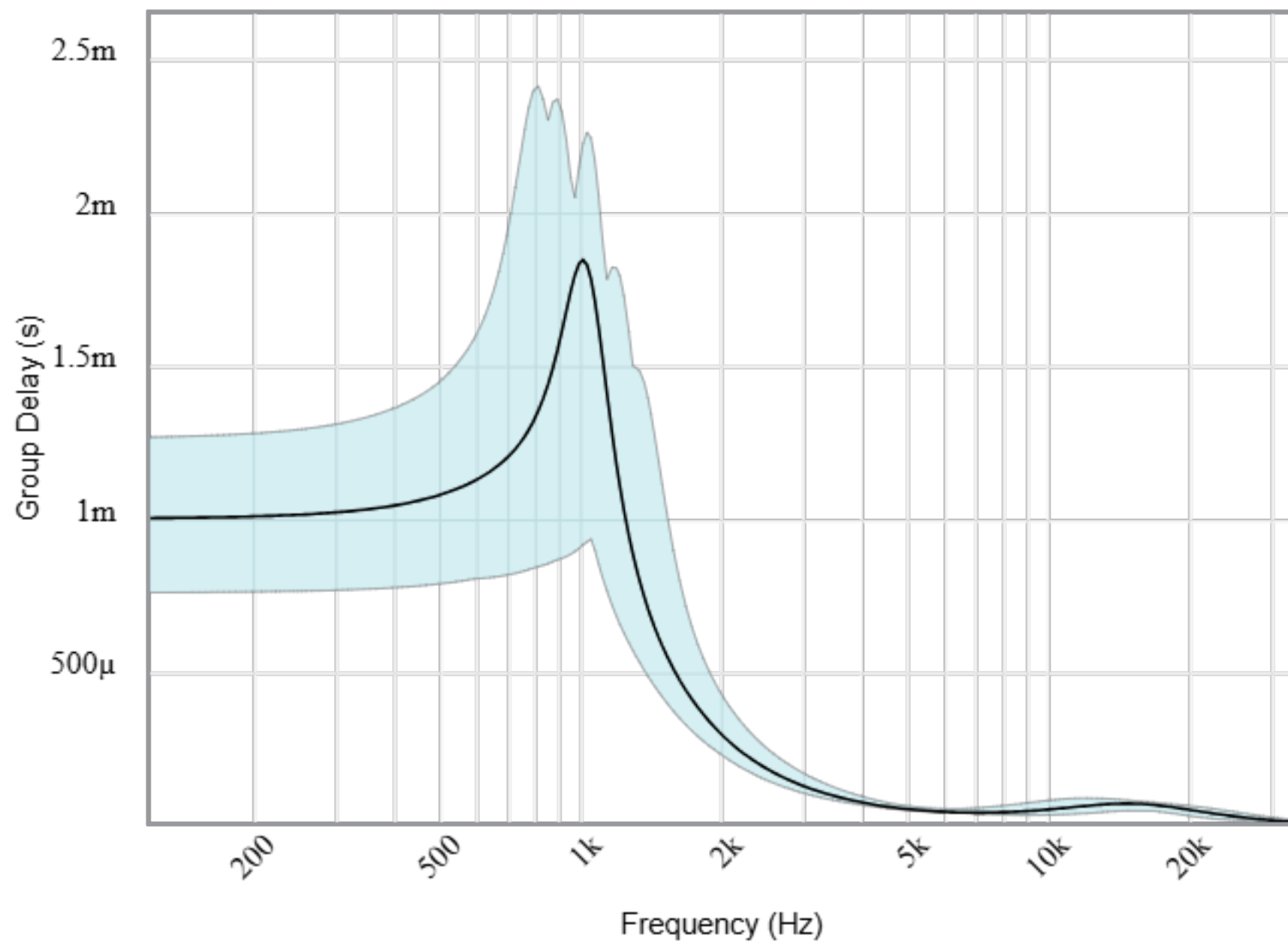
Phase(degrees)



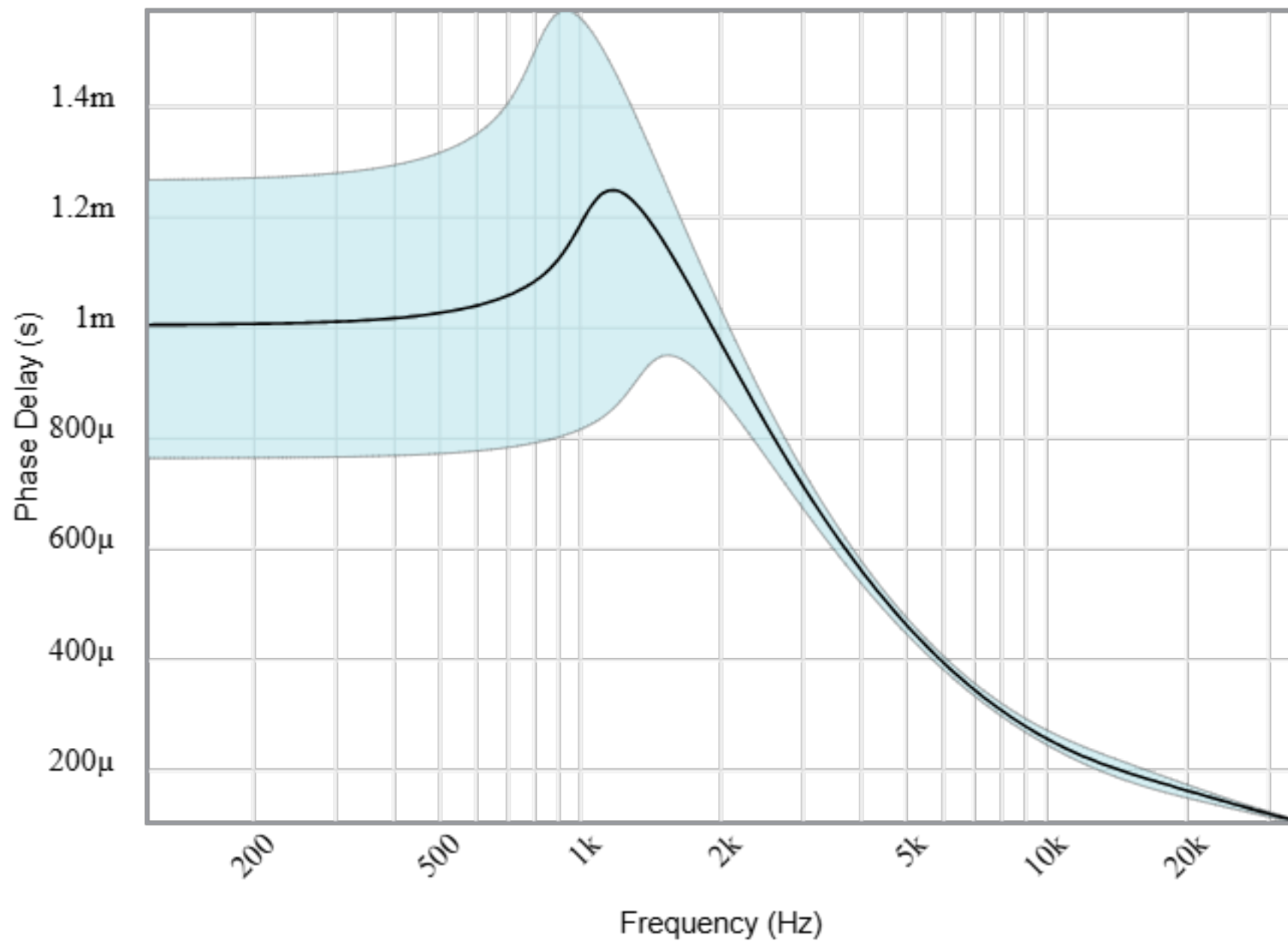
Phase(radians)



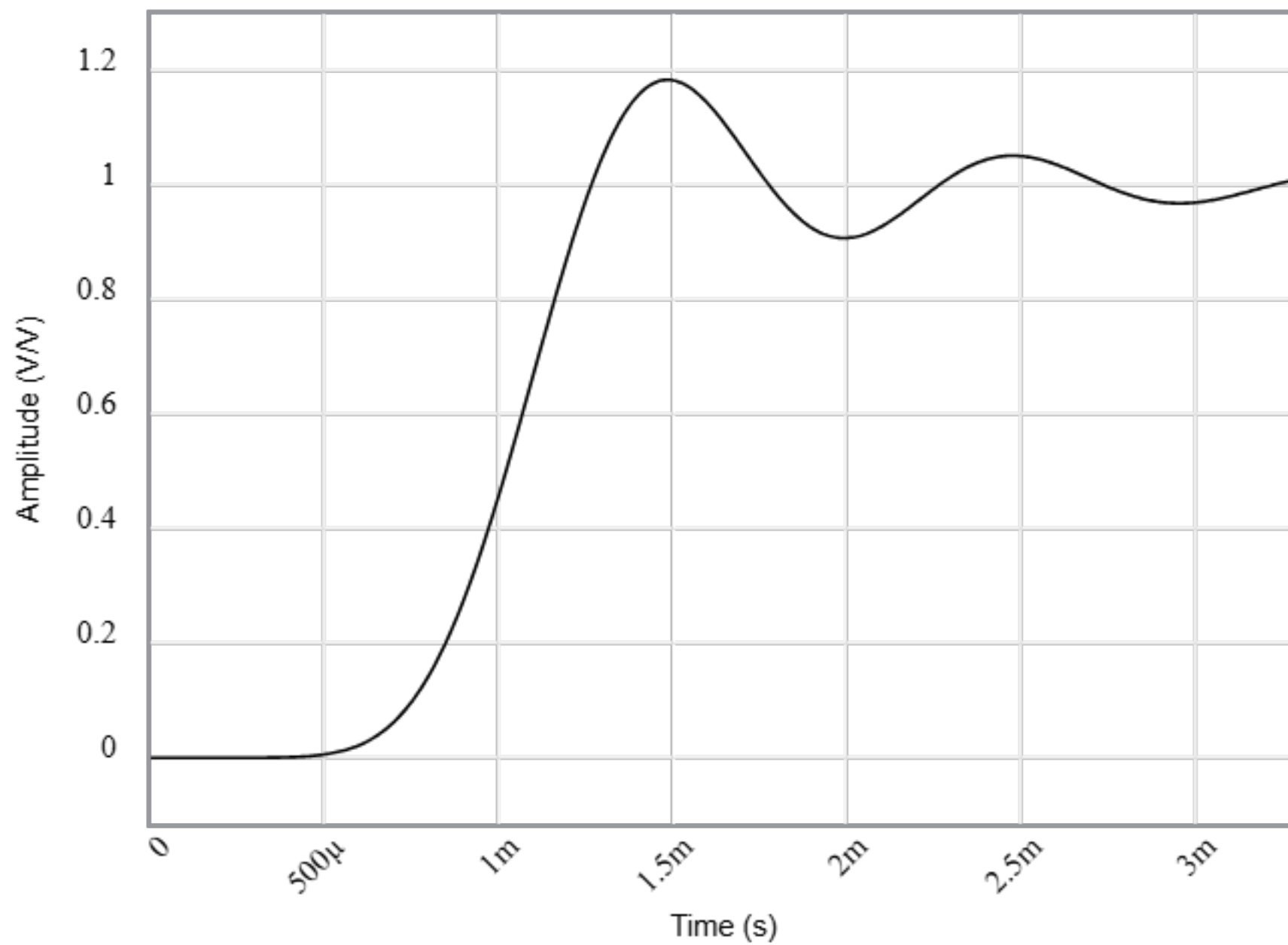
Group Delay



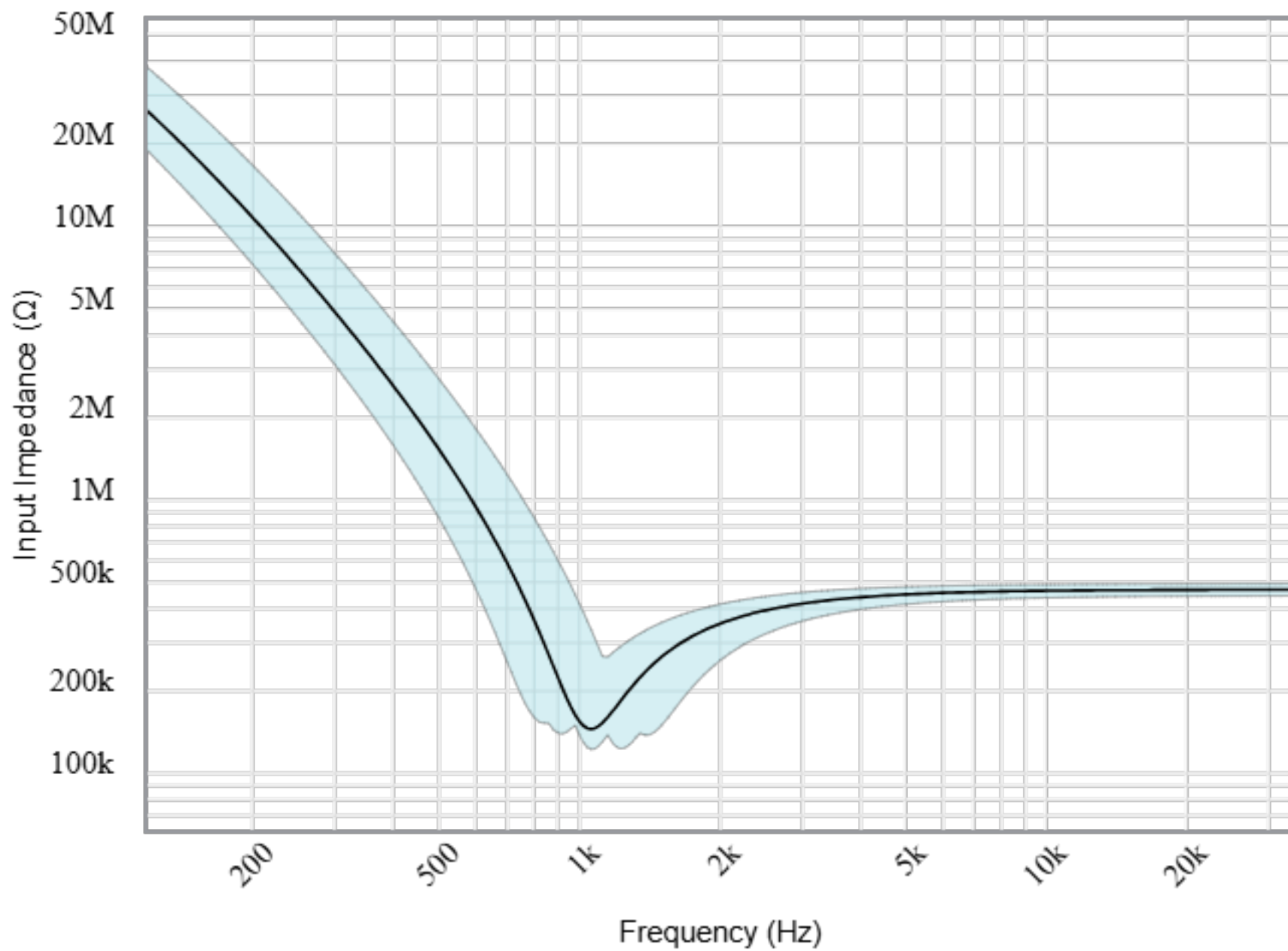
Phase Delay



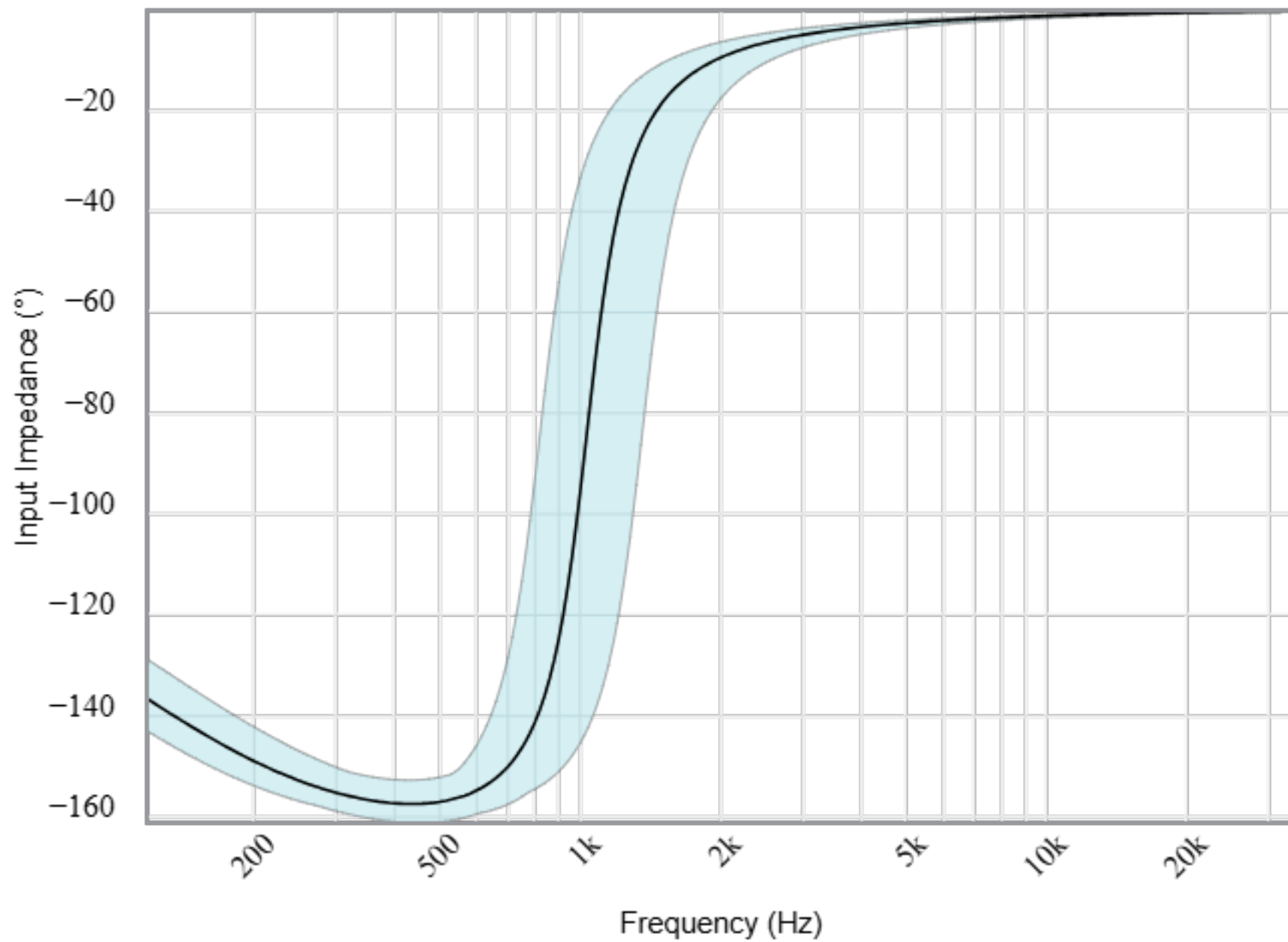
Step Response



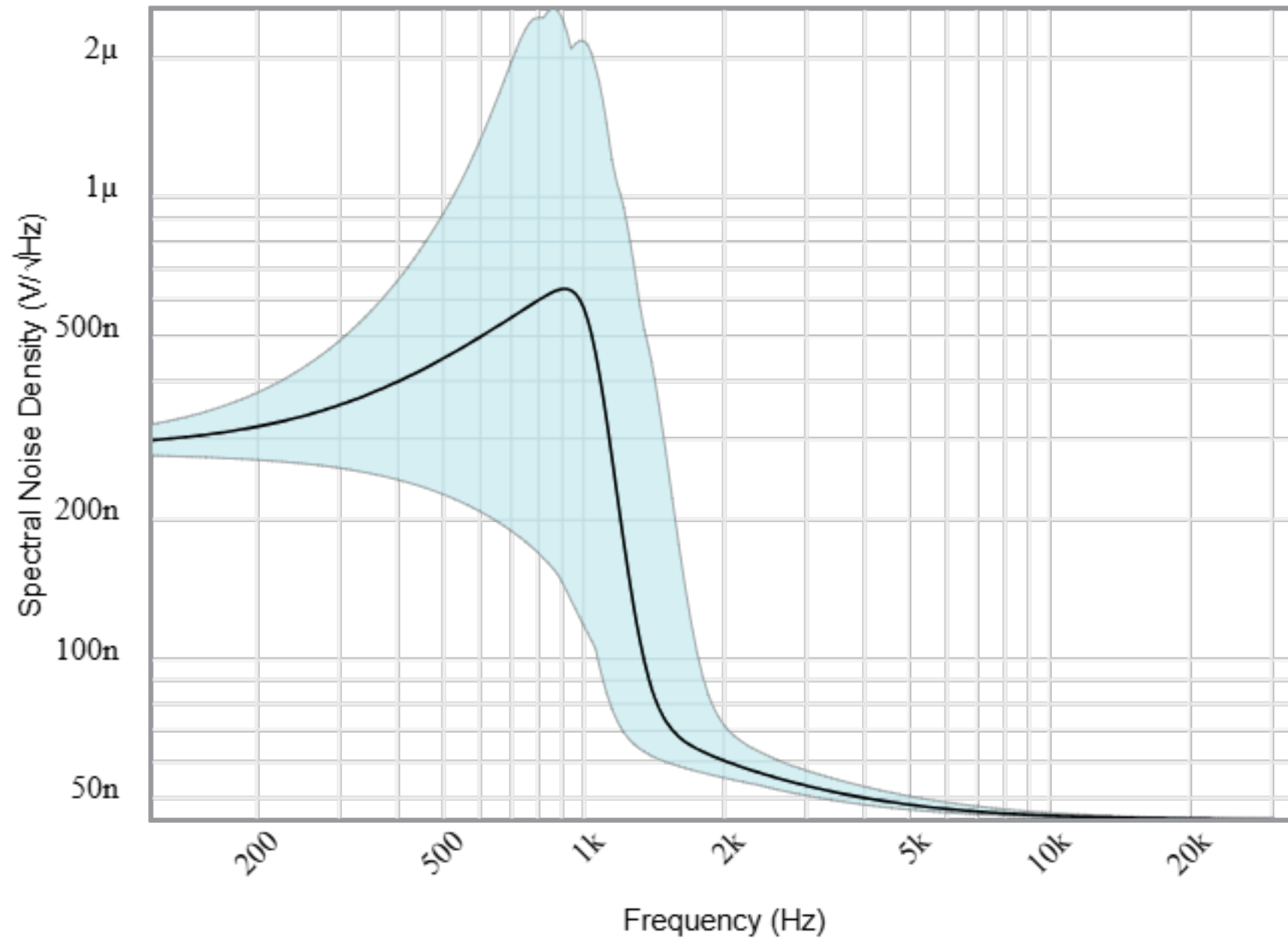
Input Impedance Magnitude



Input Impedance Phase



Noise



Stages

Your filter requires 5 op amp stage(s) with the following characteristics

A 2nd order Low-Pass Sallen Key

	Target	Simulated
Gain (V/V):	1	1 to 1
f_p (Hz):	1.08k	820 to 1.36k
Q:	3.2	2.56 to 3.98

B 2nd order Low-Pass Sallen Key

	Target	Simulated
Gain (V/V):	1	1 to 1
f_p (Hz):	1.08k	804 to 1.33k
Q:	1.1	794m to 1.24

C 2nd order Low-Pass Sallen Key

	Target	Simulated
Gain (V/V):	1	1 to 1
f_p (Hz):	1.08k	840 to 1.39k
Q:	707m	553m to 862m

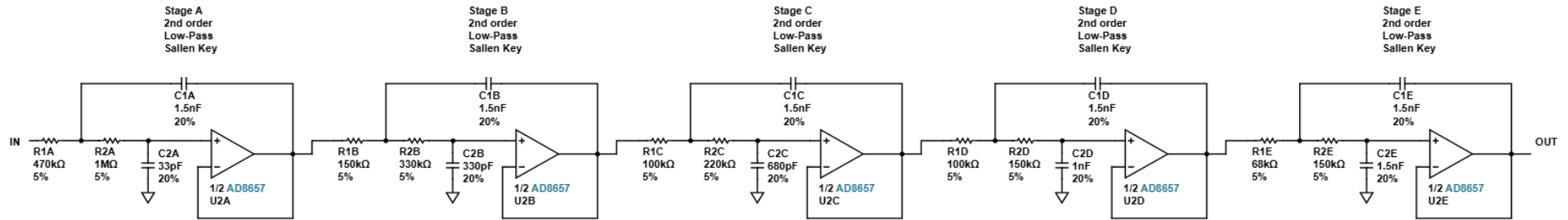
D 2nd order Low-Pass Sallen Key

	Target	Simulated
Gain (V/V):	1	1 to 1
f_p (Hz):	1.08k	839 to 1.39k
Q:	561m	486m to 745m

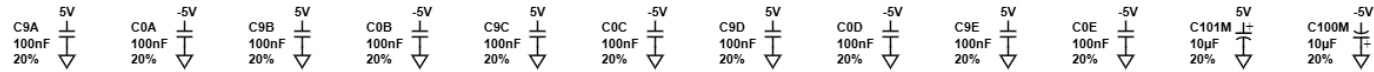
E 2nd order Low-Pass Sallen Key

	Target	Simulated
Gain (V/V):	1	1 to 1
f_p (Hz):	1.08k	830 to 1.37k
Q:	506m	373m to 580m

Circuit



BYPASS CAPACITORS



SPARES Why The Spares?

